

# Centering Harm in Socio-technical Systems: Connecting Design, User Experience, and Policy

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## Abstract

Technology-related harms are increasingly understood as systemic issues that unfold across users' interactions with digital systems. These harms ranging from subtle manipulation to psychological distress highlight the need to examine how design choices contribute to their emergence and how policy mechanisms respond. Although HCI and public policy both aim to support user well-being, they often operate in silos, limiting accountability and reducing opportunities for effective intervention. A key challenge is the social-technical gap: the persistent tension between social needs and technical constraints that hinders holistic responses. As regulatory attention grows around manipulative design practices, it is critical to develop an integrated understanding of harm- one that centers it within sociotechnical systems by linking design, user experience, and policy. My research unfolds in two parts: first, I investigate how harm is constructed, perceived, and regulated; second, I explore how HCI and policy scholars can foster a bi-directional relationship to collaboratively shape ethical technologies and responsive regulation.

## CCS Concepts

• **Social and professional topics** → **Computing / technology policy**; • **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → **Law**.

## Keywords

sociotechnical systems, knowledge complexity, harm, ethics, policy, UX

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## 1 Introduction

Design is inherently complex, shaped not only by task and context but also by the expertise and positionality of the designer [29]. This

complexity deepens when layered with ethical and sociotechnical concerns—what scholars have termed ethical complexity [9] and knowledge complexity [10]. In HCI, its inherently interdisciplinary and volatile nature further complicates the efforts to define, prioritize, and address evolving concerns [3, 5]. Despite a growing emphasis on values such as ethical design, privacy, and fairness, there remains little consensus on which perspectives UX should center [19]. As Liu et al. [20] note, “*no tradition is the only tradition*”; in this unsettled design space, even the core concepts of UX are continually contested [18]. At the heart of this complexity lies the socio-technical gap: the persistent mismatch between nuanced human activity and the technical mechanisms available to support it, as articulated by Ackerman [1], highlighting that “*human activity is highly nuanced and contextualized*”. To effectively address this gap, it is crucial to understand how users perceive technology design and how designers' views on ethics, privacy, and manipulation significantly shape user experiences [30, 36]. Growing regulatory action against deceptive and manipulative design practices has further emphasized the designer's role and their ethical responsibility.

My research engages directly with this challenge by centering *harm* as a shared concern across design, user experience, and policy. I aim to connect designer intentions, lived user experiences, and regulatory outcomes to foster a more integrated understanding of harm. Through this work, I seek to strengthen collaboration between HCI and public policy, fostering intentional, cross-disciplinary approaches to mitigating digital harms and prioritizing user well-being. Specifically, I explore how insights from both designers and users can inform the development of more ethically grounded and responsive policy frameworks.

### 1.1 Preliminary Work

**1.1.1 Engagement with (design) complexity.** : In Gray et al. [10], we expanded the concept of design complexity to include knowledge complexity, which involves navigating multiple, often conflicting, forms of knowledge within UX practice. Through semi-structured interviews and case studies, we explored how UX practitioners engage with legal and regulatory frameworks in their work. Our findings revealed significant variation in how these frameworks are understood and applied, influenced by factors like educational background, organizational culture, and personal beliefs. For some, legal knowledge aligned with business goals; for others, it was seen as a constraint on creativity. These tensions highlight the challenges of integrating legal perspectives into design processes. My ongoing research (including an in-progress article which will be submitted to TOCHI) extends these findings by exploring how different knowledge types—ethical, technical, organizational, and

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legal—interact within design decision-making. This work identifies how designers navigate trade-offs and reveals points where collaborative frameworks can help align design practice with ethical and policy goals.

**1.1.2 Learning from the regulation against dark patterns.** : Digital harm often stems not from malfunction, but from intentional design choices that exploit attention and undermine user agency [22]. Harry Brignull, the originator of the term, emphasized that dark patterns span entire user journeys, a dynamic process further explored by Gray et al. [8]. Our research [11] introduces a visually-focused methodology to track these deceptive tactics over time. Analyzing documentation from a pending US Federal Trade Commission lawsuit against Amazon, we focused on the subscription flow and propose three levels for analyzing dark patterns: Intra-page, Inter-page, and System level. This framework, *Temporal Analysis of Dark Patterns* (TADP), offers a systematic approach for characterizing manipulative practices and describing the harms they cause. This methodology can be used by designers, researchers, and regulators to create ethically sound, user-centric designs. The shift from practitioner-driven critique to regulatory recognition of dark patterns underscores the need for standardized terminologies and frameworks to assess and mitigate harm effectively. This work aligns with my broader interest in understanding that harms are not static, but persist across user experiences [4, 23], and underscores the importance of my goal to create shared resources for designers, scholars, and policymakers to collaboratively address these issues.

**1.1.3 Locating intersections between HCI and public policy.** : The HCI community has long recognized its potential to influence policy, with early discussions dating back to 1998 [13]. By 2013, scholars began critically examining how CSCW and HCI could inform science and technology policy [16]. The conversation deepened in 2014 with the *knot model* [15], which emphasized the interconnectedness of policy, practice, and design in shaping social computing systems. By 2024, HCI's role in policy had become both influential and necessary, with growing calls to integrate policy considerations directly into HCI research [35]. In our systematic review [7] of publications at CHI, DIS, and CSCW, we found growing recognition of public policy across HCI, but a clear gap remains in translating that awareness into design outcomes. We constructed three core themes that bridge HCI and policy, and illustrate how policy's engagement in HCI discussions range from passive background influence, to working at the intersection, to becoming an equal partner in the foreground. While few HCI scholars have directly embedded legal frameworks into design [12, 17, 34], we offer a provocation for future work: imagining a space where HCI can shape, inform, and co-create policy, while also ensuring that design remains ethically and legally grounded.

## 2 Research Focus

My research aims to explore the interdisciplinary complexity of HCI and policy, focusing on how technology-related harms emerge and persist across the user experience. These harms, even when unintentional, can result from the complex interactions between design decisions, user experiences, and regulatory frameworks.

Building on my previous work on design complexity and dark patterns, I plan to address the following research questions:

- (1) How is the concept of “harm” constructed, perceived, and addressed in HCI scholarship?
- (2) How we can integrate a pluralistic understanding of harm into both technology and policy design, encouraging collaboration between HCI scholars and policymakers to develop more effective, ethically grounded harm mitigation strategies?

## 3 Part 1: Centering Harm

Harm is a complex concept shaped by society and law, involving judgments about its severity and consequences and scholars have called to look beyond technical boundaries in order to map its complexities [28]. Prior scholarship has developed taxonomies and models to capture the nuances of harm—including a typology of harm and severity [25], a taxonomy of cyber-harm [2], and a legally-grounded harm taxonomy [24]. Harm is understood through diverse lenses and conceptual frameworks, with related notions like safety [33], online risk [6], trust [32], and misinformation [27], and harms of emerging technology like deepfakes [31] and virtual reality [26] offering timely insights into the kinds of impacts that can result from emerging technologies. Acknowledging Ackerman [1]'s social-technical gap, this work centers harm as both a systemic and lived experience, seeking to understand it from the perspectives of designers, researchers, users, and policymakers.

- **Study 1: Harm as a design consideration [UX Practice] (planning):** Designers face growing complexity in addressing ethical harms, particularly as AI introduces new forms of deceptive practices. This study investigates how UX practitioners perceive and prioritize ethical harm within their design work, especially when legal and business interests are in tension. In this study, I investigate how UX practitioners identify, interpret, and address ethical harms in real-world design contexts by reflecting on decision-making scenarios in co-design workshops, guided by a harm taxonomy [24], with the aim of grounding ethical frameworks in practice and supporting responsible, legally informed outcomes.
- **Study 2: Harm as a design consideration [HCI Research] (data collection):** This study systematically reviews how HCI and design researchers construct scenarios to assess harm caused by emerging technologies. By examining existing literature, I aim to understand how scenario-based approaches can be applied to identify, interpret, and mitigate potential harms, particularly in the context of AI and deceptive practices. The goal is to develop a set of best practices for scenario construction that can guide both researchers and practitioners in addressing ethical challenges. This work will inform policymakers and designers, offering actionable insights into understanding and mitigating the harms of emerging technologies.
- **Study 3: Harm as a felt experience [Specific Contexts] (data collection):** While dark patterns in games and social media are well-documented, less is known about how users actually experience and interpret these manipulative designs. In this study, I ask: *How do teenagers perceive and describe*

*harm in the digital environments they use most—particularly video games and social media?* Using interviews and participatory activities, we will gather teens' narratives about dark patterns and co-develop ideas for design and policy interventions. Focusing on teens' felt experiences allows us to connect ethical and regulatory frameworks to user-centered evidence, supporting youth-informed design guidance and grounded policy development that enhances digital well being.

- **Study 4: Harm as a policy challenge [Law and Regulation] (planning):** Digital harm is increasingly addressed through legal and regulatory frameworks, yet the definition and application of "harm" remains inconsistent across contexts. In this study, I investigate: *How is harm operationalized in policy and governance frameworks related to digital systems?* Using document analysis, I will examine policy frameworks to understand how harm is framed and enforced across jurisdictions. This work explores the interplay between design practices and regulatory responses, examining how policy frameworks interpret and respond to design-driven harms and how these frameworks, in turn, influence design decisions.

## 4 Part 2: Bi-directional Relationships between HCI and Policy

This part builds on the understanding that technology and policy exist in a deeply interactional and optimistically bidirectional relationship neither operates in isolation, but instead they continually shape and reshape one another. Rather than viewing policy as an afterthought to technological development, this perspective positions it as an integral, co-evolving part of the design process. Echoing the approach of initiatives like the Tech Policy Lab [14, 21], case studies and design interventions that adopt this lens provide training to researchers and practitioners, enabling them to recognize how design decisions can embed or resist policy logics. Similarly, this approach illustrates how regulatory frameworks, in turn, influence technological practices. This orientation sets the stage for Study 5, which investigates the HCI–policy nexus through the lens of digital harm, responding to the need for collaborative, cross-disciplinary methods that connect regulatory insight with user-centered design.

- **Study 5: Harm at the HCI–policy nexus (planning):** As digital harm becomes a policy priority, there is a growing need to bridge HCI methods with regulatory practices. HCI researchers identify tech-related harms, and policymakers try to address them, but they don't collaborate closely, leaving a gap between research and regulation. This study explores how HCI scholars and designers can serve as a sensor for harm detection and how policy can function as a tool for accountability. What HCI and design methods can policymakers leverage to systematically identify, assess, and mitigate the harms stemming from emerging technologies?

## 5 Conclusion

Through this proposed dissertation work, I seek to center harm as a shared concern across design, user experience, and policy. Based on the expected outcomes across five studies (two currently in

progress), I will connect designer intentions, lived user experiences, and regulatory outcomes to foster a more integrated understanding of harm—bridging the gap between HCI and policy by fostering a bidirectional relationship between the two fields.

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